microCT imaging of threespine stickleback

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Abstract

Introduction

- Embedded into Alaska Stickleback Restoration Project, Genomics axis where Katie Peichel, Ben Sulser and Sheila Christen are affiliated.

Materials & Methods

Sample preparation

microtomographic imaging

- Scanned on a Bruker SkyScan 2214
- Sample holder generated with OpenSCAD, available online at GitHub [1]
  - Scanning several fish toghether to efficiently use machine time
  - Full in Bruker workflow
  - Results in PNG stacks on disk

Data analysis

- Jupyter notebooks [2]
  - Efficiently loading data from disk with `dask` [3]
  - Extract position of single fish (all scanned together), based on the MIP of the scan
  - Crop out each fish (with a buffer) and write to cropped dataset
    - Cropped datasets are saved to discrete folders for easy handling. In both original gray-scale plus as thresholded dataset, e.g. binarized into bone and “not bone”. These sare saved out as `zarr` [4] and `nrrd` files.

Results

Discussion

Conclusion

Author Contributions

Contributor Roles Taxonomy (CRediT), as defined in [5]:

- Data curation: David Haberthür
- Formal analysis: David Haberthür
- Investigation: David Haberthür
- Methodology: David Haberthür
- Project administration: David Haberthür
- Software: David Haberthür
- Validation: David Haberthür
- Visualization: David Haberthür
- Writing – original draft: David Haberthür
- Writing – review & editing: David Haberthür

Competing Interest

Author	Competing Interests	Last Reviewed
David Haberthür	None	2026-01-14
Ben Sulser		
Sheila Christen		
Katie Peichel		
Ruslan Hlushchuk		

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